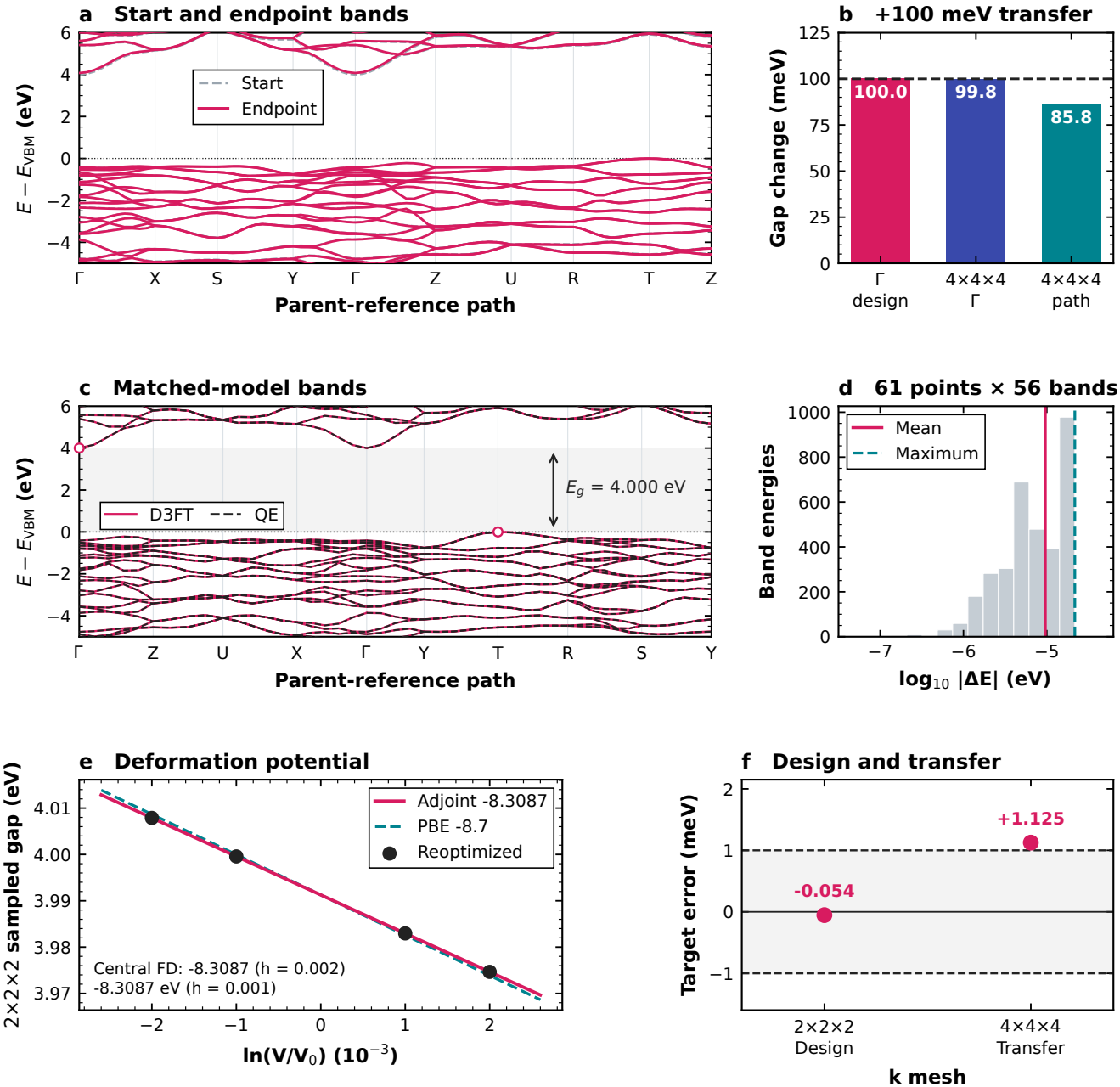

a Start and endpoint bands
Start
Endpoint
$E - E_{VBM}$ (eV)
Γ X S Y Γ Z U R T Z
Parent-reference path
b +100 meV transfer
Gap change (meV)
100.0
99.8
85.8
Γ design
4×4×4 Γ
4×4×4 path
c Matched-model bands
D3FT QE
E_g = 4.000 eV
$E - E_{VBM}$ (eV)
Γ Z U X Γ Y T R S Y
Parent-reference path
d 61 points × 56 bands
Mean
Maximum
Band energies
$\log_{10}$ |ΔE| (eV)
e Deformation potential
Adjoint -8.3087
PBE -8.7
Reoptimized
Central FD: -8.3087 (h = 0.002)
-8.3087 eV (h = 0.001)
2×2×2 sampled gap (eV)
ln(V/V$_0$) (10^{-3})
f Design and transfer
+1.125
-0.054
Target error (meV)
2×2×2 Design
4×4×4 Transfer
k mesh